

PROGRAM 1

OUTPUT

```
[ita1@Kgcaslinux ~]$ ls
file1 file2 sun
[ita1@Kgcaslinux ~]$ cp file1 file3
[ita1@Kgcaslinux ~]$ cat file3
a b c d e f g
[ita1@Kgcaslinux ~]$ mv file2 file4
[ita1@Kgcaslinux ~]$ wc file4
7 7 28 file4
[ita1@Kgcaslinux ~]$ split -3 file4
[ita1@Kgcaslinux ~]$ ls
file1 file3 file4 sun xaa xab xac
[ita1@Kgcaslinux ~]$ cmp file1 file4
file1 file4 differ: byte 1, line 1
[ita1@Kgcaslinux ~]$ diff file1 file4
1,7c1,7
< a < b < c ...
---
> sun > mon > tue ...
[ita1@Kgcaslinux ~]$ rm xac
[ita1@Kgcaslinux ~]$ rm -r *
```

PROGRAM 2

OUTPUT

```
[ita1@Kgcaslinux ~]$ sh sysinfo.sh
Username : tsa
Login Name : tsa
Current Shell : /bin/bash
Home Directory : /home/tsa
OS Type : linux-gnu
Current Path : /usr/local/bin:/usr/bin:/bin
Working Dir : /home/tsa
Logged-in Users : 2
Available Shells:
/bin/sh /bin/bash /usr/bin/zsh
CPU Info :
Model name: Intel(R) Core(TM) i5-8250U CPU @ 1.60GHz
Memory Info :
total used free
Mem: 7892 3400 4492
```

PROGRAM 3

OUTPUT

```
[tsa03@localhost ~]$ sh pipe_redirect.sh --- Pipe: Count files using ls | wc ---
64 569 3333
--- Redirection (>): Write to file ---Welcome to Linux
--- Append (>>): Add to file ---
Welcome to Linux
Always best learning institute
--- More: Display files page by page ---total 504
drwx----- 4 tsa03 tsa03 4096 Feb 13 08:40 .
-rw-rw-r-- 1 tsa03 tsa03 24 Jan 23 aa
--- Tee: Screen + Save to file ---
-rw-r--r-- 1 tsa03 tsa03 24 Jun 3 sample1.txt Content saved in tee_output.txt:
-rw-r--r-- 1 tsa03 tsa03 24 Jun 3 sample1.txt
```

PROGRAM 4

OUTPUT

```
[ita1@Kgcaslinux ~]$ sh menu.sh
1- Who am I?
2- Who is logged on?
3- Date
4- Calendar
5- Current Directory
6- File Listing
Enter your choice:
2
tsa pts/0 Jun 3 09:00 (192.168.1.10)
Enter your choice:
3
Mon Jun 3 09:15:30 IST 2026
```

PROGRAM 5

OUTPUT

```
[cs2ax@kgcaslinux ~]$ sh filter.sh
Filter Commands Demo
Lines, words and characters in passwd:
1404 1451 65016 passwd
Lines containing the word root:
1:root:x:0:0:root:/root:/bin/bash
10:operator:x:11:0:operator:/root:/sbin/nologin
Count of lines containing root:
2
Lines that do NOT contain root:
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
Replace ':' with '*' in passwd:
root*x*0*0*root*/root*/bin/bash
First column (usernames) of passwd:
root
bin
Daemon
```

PROGRAM 6

OUTPUT

```
[cs2ax@kgcaslinux ~]$ sh zero_file.sh
Enter any file name:
one
one file exists
one file has size > 0
[cs2ax@kgcaslinux ~]$ sh zero_file.sh
Enter any file name:
empty
empty file exists
empty deleted (size was 0 bytes)
[cs2ax@kgcaslinux ~]$ sh zero_file.sh
Enter any file name:
second
second file does not exist
```

PROGRAM 7

OUTPUT

```
[cs2ax@kgcaslinux ~]$ sh digit_sum.sh
Enter a number: 12
Sum of all digits is 3
[cs2ax@kgcaslinux ~]$ sh digit_sum.sh
Enter a number: 4567
Sum of all digits is 22
```

PROGRAM 8

OUTPUT

```
[ctb11@Kgcaslinux ~]$ sh greatest.sh 3
```

```
Enter number 1:
```

```
3
```

```
Enter number 2:
```

```
98
```

```
Enter number 3:
```

```
56
```

```
Numbers entered are:
```

```
3 98 56
```

```
Smallest number in the array: 3
```

```
Greatest number in the array: 98
```

PROGRAM 9

OUTPUT

```
[ctb11@Kgcaslinux ~]$ sh palindrome.sh
```

```
Enter Number: 121
```

```
Number 121 is a Palindrome
```

```
[ctb11@Kgcaslinux ~]$ sh palindrome.sh
```

```
Enter Number: 1234
```

```
Number 1234 is NOT a Palindrome
```

PROGRAM 10

OUTPUT

```
[csb1@Kgcaslinux ~]$ sh multiplication.sh
Enter the table number:
5
Enter the range:
10
Multiplication table for 5 up to range 10
1 X 5 = 5
2 X 5 = 10
3 X 5 = 15
4 X 5 = 20
5 X 5 = 25
6 X 5 = 30
7 X 5 = 35
8 X 5 = 40
9 X 5 = 45
10 X 5 = 50
```

PROGRAM 11

OUTPUT

```
[csb1@Kgcaslinux ~]$ sh group_info.sh
--- All Groups in the System ---
root
bin
daemon
users ...
Enter group name to find its ID:
users
Group ID of 'users' is: 100
Enter group name to list its members:
users
Users in group 'users':
csb1
csb2
Total number of groups: 72
Groups assigned to current user (csb1):
csb1 users wheel
```

PROGRAM 12

OUTPUT

```
[csb1@Kgcaslinux ~]$ sh process_monitor.sh
--- All Running Processes ---
UID PID PPID C STIME TTY TIME CMD
root 1 0 0 09:00 ? 00:00:02 /sbin/init
root 2 0 0 09:00 ? 00:00:00 [kthreadd] ...
--- Processes of Current User (csb1) ---
PID TTY TIME CMD
1234 pts/0 00:00:00 bash
--- Top 5 CPU Consuming Processes ---
PID COMMAND %CPU
1024 firefox 12.5
1100 Xorg 5.2
1200 python3 3.1
1300 bash 0.5
1400 ps 0.2
--- PID and PPID ---
PID PPID COMMAND
1 0 systemd
2 0 kthreadd
Total number of running processes: 185
```

PROGRAM 13

OUTPUT

```
[csb1@Kgcaslinux ~]$ sh network_info.sh
--- System Hostname ---
Kgcaslinux
--- IP Addresses of Network Interfaces ---
1: lo: inet 127.0.0.1/8 scope host lo
2: eth0: inet 192.168.1.100/24 brd 192.168.1.255
--- Routing Table ---
default via 192.168.1.1 dev eth0
192.168.1.0/24 dev eth0 proto kernel
--- DNS Server Configuration ---
nameserver 8.8.8.8
nameserver 8.8.4.4
Enter remote host to test (e.g. 8.8.8.8):
8.8.8.8
Testing connectivity to 8.8.8.8 ...
64 bytes from 8.8.8.8: icmp_seq=1 ttl=118 time=12.3 ms
4 packets transmitted, 4 received, 0% packet loss
```

PROGRAM 14

OUTPUT

```
[csb1@Kgcaslinux ~]$ sh log_monitor.sh
--- Recent System Log Entries ---
Jun 3 09:00:01 Kgcaslinux systemd: Started Session 5 of user csb1. Jun 3
09:01:15 Kgcaslinux kernel: eth0: link is up ...
--- Login History of Users ---
csb1 pts/0 192.168.1.10 Mon Jun 3 09:00 still logged in
root pts/1 192.168.1.5 Mon Jun 3 08:45 - 08:55 (00:10)
--- Failed Login Attempts ---
Jun 3 08:30 sshd: Failed password for invalid user admin from 10.0.0.1 Enter
keyword to search in /var/log/messages:
error
Results for 'error':
Jun 3 09:02 kernel: ACPI Error: Method parse/execution failed
--- Last 10 Security Log Messages ---
Jun 3 09:00 sshd: Accepted password for csb1 from 192.168.1.10
```

PROGRAM 15

OUTPUT

```
[csb1@Kgcaslinux ~]$ sh disk_usage.sh
--- Disk Usage of Home Directory ---
4.0K /home/csb1/Desktop
8.5M /home/csb1/Documents
120M /home/csb1/Downloads
--- Top 5 Largest Directories ---
12G /usr
5.2G /var
3.8G /home
1.5G /opt
980M /boot
--- File Count in Each Home Subdirectory ---
/home/csb1/Desktop/ : 3 files
/home/csb1/Documents/ : 12 files
/home/csb1/Downloads/ : 45 files
--- Filesystem Type of Each Partition ---
Filesystem Type Mounted on
/dev/sda1 ext4 /
tmpfs tmpfs /tmp
--- Free Disk Space Available ---
Filesystem Size Used Avail Use% Mounted on
/dev/sda1 50G 22G 26G 46% /
tmpfs 3.9G 0 3.9G 0% /tmp
```